

4.12.13

AI25BTECH11008 - Chiruvella Harshith Sharan

Question:

If the equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is $(2, -1)$, then find the length of the side of the triangle. Also find the other vertices.

Answer:

Step 1: Represent line in matrix form

The line equation:

$$x + y = 2 \implies \mathbf{n}^T \mathbf{x} = 2,$$

where

$$\mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Vertex:

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Step 2: Foot of perpendicular (matrix equation)

The foot $\mathbf{D}$ is obtained by projecting $\mathbf{A}$ onto the line:

$$\mathbf{D} = \mathbf{A} - \frac{\mathbf{n}^T \mathbf{A} - 2}{\mathbf{n}^T \mathbf{n}} \mathbf{n}.$$

Substitute:

$$\mathbf{D} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} - \frac{(1)(2) + (1)(-1) - 2}{1^2 + 1^2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

Step 3: Side length of equilateral triangle

In an equilateral triangle, altitude $AD = \frac{\sqrt{3}}{2}a$.

Compute:

$$AD = \|\mathbf{A} - \mathbf{D}\| = \sqrt{\left(2 - \frac{5}{2}\right)^2 + \left(-1 + \frac{1}{2}\right)^2} = \frac{1}{\sqrt{2}}.$$

So:

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2}a \implies a = \sqrt{\frac{2}{3}}.$$

Step 4: Coordinates of other vertices

Let $\mathbf{m}$ be a direction vector along the base line. Since $\mathbf{n} = (1, 1)^T$, take

$$\mathbf{m} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Half the base length is $\frac{a}{2}$. So, the base vertices are:

$$\mathbf{B} = \mathbf{D} + \frac{a}{2} \frac{\mathbf{m}}{\|\mathbf{m}\|}, \quad \mathbf{C} = \mathbf{D} - \frac{a}{2} \frac{\mathbf{m}}{\|\mathbf{m}\|}.$$

Since $\|\mathbf{m}\| = \sqrt{2}$:

$$\mathbf{B} = \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \end{pmatrix} + \frac{\sqrt{2/3}}{2\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} + \frac{1}{2\sqrt{3}} \\ -\frac{1}{2} - \frac{1}{2\sqrt{3}} \end{pmatrix},$$

$$\mathbf{C} = \begin{pmatrix} \frac{5}{2} \\ -\frac{1}{2} \end{pmatrix} - \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} - \frac{1}{2\sqrt{3}} \\ -\frac{1}{2} + \frac{1}{2\sqrt{3}} \end{pmatrix}.$$

Final Answer:

$$\text{Side length } a = \sqrt{\frac{2}{3}}, \quad \mathbf{B} = \left(\frac{5}{2} + \frac{1}{2\sqrt{3}}, -\frac{1}{2} - \frac{1}{2\sqrt{3}} \right), \quad \mathbf{C} = \left(\frac{5}{2} - \frac{1}{2\sqrt{3}}, -\frac{1}{2} + \frac{1}{2\sqrt{3}} \right).$$

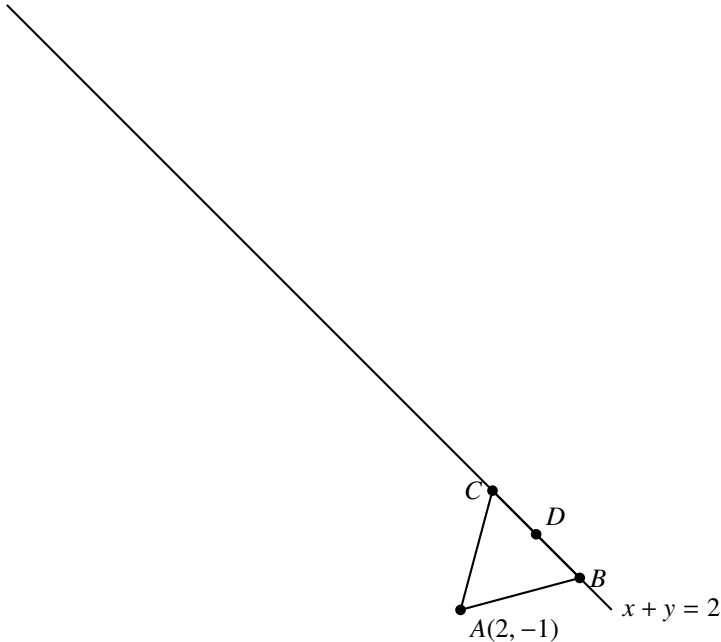


Fig. 0.1: Equilateral triangle with base on $x + y = 2$